
When Averages Lie: What 38 Million Patient Records Reveal About Multi-Site COVID-19 Studies

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Abstract

Large multi-site medical studies pool data from dozens of hospitals to draw conclusions about disease. A hidden assumption in this pooling is that hospital-level averages faithfully represent what happens to individual patients. We test this assumption directly. Using 38 million individual COVID-19 patient records from the CDC and 1,540 records from two New York hospitals, we show that a standard site-level analysis cannot even detect the relationship between age and mortality ($p = 0.12$), while individual-level analysis finds that patients over 80 have 9.9 times the odds of dying ($p < 10^{-300}$). This is the ecological fallacy in action: what holds for groups does not hold for people. We then show that structural consistency tests—methods that examine the *structure* of relationships across sites, rather than averaging over them—can detect this kind of hidden heterogeneity even from site-level data alone. Applied to the 4CE consortium’s 22,000-patient neurological COVID-19 dataset, these methods reveal systematic patterns that standard meta-analysis misses entirely. The practical message: when multi-site studies report a pooled average, readers should ask whether the average could be hiding the opposite conclusion at the patient level.

1 Introduction: The Problem with Averages

Suppose ten hospitals each report the fraction of their COVID-19 patients who died. A meta-analysis averages those fractions and reports the result as “the” mortality rate. This is the standard approach, and it works when the hospitals are measuring the same thing in the same kind of patient.

They rarely are.

Hospitals differ in who walks through the door. One serves a young urban population; another, a rural elderly community. One codes aggressively for neurological complications; another barely records them. The average across these hospitals is a real number, but it may not describe any actual hospital or any actual patient. Worse, it can point in the wrong direction entirely.

This problem—drawing conclusions about individuals from group-level data—has a name: the *ecological fallacy* Robinson (1950). It has been known since 1950, taught in every epidemiology course, and routinely ignored in practice. Multi-site consortium studies, including the largest COVID-19 analyses, typically have access only to site-level summaries and treat them as if they represent patient-level truths.

We set out to test how badly this matters. We obtained individual patient records from two public sources—38 million cases from the CDC Case Surveillance dataset and 1,540 patients from two New York hospitals—and compared what you learn from site-level analysis versus what you learn from patient-level analysis of the same data. The gap is dramatic.

We then ask: if you are stuck with site-level data (as most consortia are), is there anything better than the standard average? We show that structural consistency tests—methods inspired by algebraic topology and differential geometry, though simpler than their namesakes—can detect structural problems in site-level data

Table 1: The ecological fallacy demonstrated. The same 38.2 million patients produce opposite conclusions depending on whether they are analyzed as groups or as individuals. Site-level analysis cannot detect the age–mortality relationship; individual-level analysis finds it with overwhelming significance.

Analysis level	What is tested	Effect size	p -value
Site-level (9 groups)	Proportion 80+ vs. mortality rate	$\beta = +0.28$	0.12
Individual (12.7M complete)	Age per decade $\rightarrow$ death (logistic)	OR = 2.40	$< 10^{-300}$
Individual (12.7M)	Age 80+ vs. not (logistic)	OR = 9.88	$< 10^{-300}$

that standard methods miss. These methods do not replace individual-level analysis, but they can warn you when the average is lying.

2 What We Found

2.1 The same data, two different answers

We start with the simplest possible question: does age predict COVID-19 mortality? The answer is obviously yes. The question is whether site-level analysis can find it.

The CDC Case Surveillance dataset contains 38.2 million confirmed COVID-19 cases with age group, sex, race/ethnicity, hospitalization status, ICU admission, and death. We used race/ethnicity categories as pseudo-sites (9 groups with $n \geq 1,000$), mimicking what a consortium study would have if each racial/ethnic group were a separate hospital reporting summary statistics.

Site-level analysis aggregated each group’s mortality rate and proportion of elderly patients, then regressed one on the other across the 9 groups. Result: no significant relationship ($\beta = +0.28$, $p = 0.12$, $R^2 = 0.31$). Site-level analysis cannot detect that age predicts mortality.

Individual-level analysis on the same 38.2 million patients found that being over 80 multiplies the odds of death by 9.9 ($p < 10^{-300}$). After adjusting for sex and comorbidities, patients gain 2.4 times the odds of dying for each additional decade of age ($p < 10^{-300}$). Males have 1.7 times the odds; patients with underlying conditions have 1.5 times the odds.

The effect is overwhelming at the individual level and invisible at the site level. This is the ecological fallacy.

2.2 Why does this happen?

The ecological fallacy is counterintuitive, so it helps to walk through the mechanism.

Consider two racial/ethnic groups. Group A has a high proportion of elderly patients *and* a high mortality rate. Group B has a low proportion of elderly patients *and* a low mortality rate. If this were the whole story, the site-level regression would work: more elderly $\rightarrow$ more death.

The problem is confounding. Groups also differ in sex ratios (ranging from 15% to 65% male), comorbidity prevalence (0.5% to 14%), access to care, and coding practices. These differences scramble the age–mortality relationship at the group level. For example, the group with the highest proportion of 80+ patients (“Unknown” race, 13.5%) has lower mortality (3.3%) than the group with the second-highest elderly proportion (Hispanic, 9.4%, mortality 7.6%), because the Unknown group has different sex composition and comorbidity rates.

At the individual level, these confounders are controlled for directly: the logistic regression asks “among patients with the same sex and comorbidity status, does age predict death?” The answer is unambiguous.

This is not a subtle statistical point. It means that any multi-site study reporting pooled averages without individual-level data could be reporting the opposite of what actually happens to patients.

2.3 The 4CE consortium: a real-world example

The 4CE consortium (Consortium for Clinical Characterization of COVID-19 by EHR) analyzed neurological outcomes in 22,000 hospitalized patients across 21 hospitals in 6 countries Chou et al. (2021). They found that central nervous system involvement roughly doubles the odds of severe disease (proportional morbidity index ≈ 2.0).

This is a site-level analysis. The consortium has access only to aggregate counts from each hospital, not individual patient records. The ecological fallacy applies directly.

How badly? We can estimate. The between-site heterogeneity is extreme: $I^2 = 99.8\%$, Cochran’s $Q = 6318$. Standard meta-analysis methods diverge by 350-fold on this data (fixed-effects PMI = 2.0 vs. random-effects PMI = 702). The heterogeneity is so severe that the choice of statistical method determines whether neurological COVID-19 doubles your risk or increases it by a factor of 700.

Meta-regression identifies the source: the proportion of elderly patients at each site is the strongest predictor of the neurological severity effect ($\beta = +14.2$, $p < 10^{-15}$). Sites with more elderly patients report dramatically larger neurological effects. This is exactly the pattern that produces ecological fallacies—site-level confounding between age composition and the outcome of interest.

Without individual patient records, the 4CE analysis cannot tell us whether:

1. Neurological COVID-19 genuinely doubles mortality (the claimed conclusion), or
2. Hospitals that happen to code more neurological diagnoses also serve older, sicker patients who would have died anyway (the ecological alternative).

Both explanations are consistent with a site-level PMI of ≈ 2.0 .

2.4 Can structural consistency tests help?

When individual data is unavailable, is there anything better than just reporting the average and hoping for the best?

We tested five approaches, spanning three levels of mathematical sophistication. The simplest test whether sites agree on single numbers; the most sophisticated test whether the geometric structure of the data differs across sites.

Level 1: Univariate heterogeneity. Cochran’s Q asks whether sites agree on a single quantity (e.g., the mortality rate). It tests one variable at a time. This is the standard meta-analysis approach.

Level 2: Structural consistency. Each site produces a correlation matrix capturing how age, mortality, comorbidity, sex, hospitalization, and ICU admission relate to one another. The multivariate consistency statistic (inspired by cellular sheaf cohomology Robinson (2017), though we compute a permutation test on correlation matrices rather than a formal coboundary operator) measures whether these matrices disagree more than expected by chance. The interaction heterogeneity test asks whether second-order effects (e.g., age $\times$ comorbidity on mortality) vary across sites.

Level 3: Geometric distance. Two methods from mathematical statistics measure how far apart sites are in a precise geometric sense. *Information geometry* treats each site as a point on a statistical manifold and computes Fisher–Rao geodesic distances—the natural metric on the space of probability distributions. *Optimal transport* measures the minimum “work” needed to reshape one site’s patient distribution into another’s (the Wasserstein distance).

Results. The geometric distance methods detected demographic heterogeneity that is *expected*: the male proportion varies most across groups (Fisher–Rao spread = 0.107), followed by elderly proportion (spread = 0.106) and comorbidity (spread = 0.031). The Black and Multiple/Other groups are farthest from the population-weighted Wasserstein barycenter. This demographic variation is real but *unsurprising*—it simply confirms that the 9 racial/ethnic groups differ in who they contain. Neither test yields a significant z -score

Table 2: Methods applied to the same 38.2 million cases, ordered by what they detect. Ecological regression misses the signal entirely. Geometric distance methods detect expected demographic variation. Structural tests have degenerate power on homogeneous-coverage data (all 9 groups report all variable pairs) but detect strong heterogeneity on the 4CE consortium data (heterogeneous coverage). Per-site interaction regressions reveal genuine variation without needing a permutation test. Individual-level regression gives the true answer.

Method	What it tests	Statistic	p -value
Ecological regression	Group proportions	$\beta = +0.28$	0.12
Fisher–Rao geodesic	Marginal distances	$z \approx 0$	0.25
Wasserstein transport	Distribution distances	$z \approx 0$	0.48
Per-site interaction	OR variation	0.23–0.69	all $< 10^{-4}$
Individual logistic	Patient-level effects	OR = 9.88	$< 10^{-300}$
<i>On 4CE (heterogeneous coverage):</i>			
Consistency statistic	Correlation structure	$z = 25$	< 0.0001

against the permutation null ($z \approx 0$), because the marginal-level differences are consistent with independent variation across variables.

The structural consistency test reveals a methodological subtlety: on the CDC data, where all 9 groups have complete observations on all 15 variable pairs, the permutation null is degenerate ($z = 0$). This occurs because within-dimension permutation preserves pairwise sums of squared differences when coverage is homogeneous. On the 4CE consortium data, where coverage is heterogeneous (10 sites report 36–296 of 1,305 possible comorbidity pairs, 10 sites report all 1,305), the same test produces $z = 25$ ($p < 0.0001$). The test’s power depends on heterogeneous coverage—a property of the data structure, not the signal.

Per-site logistic regressions reveal genuine variation in the age $\times$ comorbidity interaction across the 9 groups. All groups show that comorbidity matters less in the elderly (all OR < 1), but the magnitude varies 3-fold: from OR = 0.23 (Missing) to OR = 0.69 (Hispanic/Latino), all highly significant ($p < 10^{-4}$). The gold-standard pooled individual-level regression confirms the interaction is real (OR = 0.92 per decade, $p < 10^{-132}$). The per-site variation in this interaction is direct evidence that pooling the age–comorbidity relationship across sites loses information.

The overall picture: **sites differ in their demographic composition (expected and detected by geometric distance methods), and they differ in the relationships between demographics and outcomes (consequential for pooling and detectable by structural tests when coverage is heterogeneous)**. No single method detects both kinds of heterogeneity. The first is visible to any descriptive summary; the second requires either structural tests with heterogeneous coverage or individual-level data.

2.5 The Zenodo hospitals: a concrete cross-check

The Zenodo New York hospital dataset provides a complementary test: 1,540 patients from two hospitals (Maimonides Medical Center and SUNY Downstate), with complete demographics, outcomes, and 13 individual comorbidity flags. Two hospitals is too few for the geometric methods (which need many sites), but it demonstrates the ecological fallacy at small scale and provides a check on the individual-level analysis.

At the individual level, age increases mortality (OR = 2.65 per decade, $p < 10^{-33}$) and the hospital effect is significant after adjustment (SUNY patients have lower mortality, $p = 0.003$). The age $\times$ hospital interaction is null (OR = 0.97, $p = 0.69$), consistent with only 2 sites (the interaction needs more hospitals to be detectable).

The two hospitals differ substantially in 11 of 13 comorbidities: Maimonides has higher rates of hypertension, hyperlipidemia, and diabetes; SUNY has higher rates of asthma, COPD, and dementia. An ecological analysis

treating these two hospitals as “sites” would conflate these compositional differences with the neurological outcome, precisely the confounding we demonstrated at scale with CDC data.

3 What This Means

3.1 For the 4CE neurological findings

The 4CE consortium’s headline finding—that CNS involvement doubles COVID-19 severity—may be correct. The biological mechanisms are plausible.

The problem is that site-level evidence cannot distinguish a genuine biological effect from ecological confounding. The extreme heterogeneity ($I^2 = 99.8\%$), the 350-fold method divergence, and the strong dependence on site demographics all point to a dataset where the site-level average is unreliable.

To resolve the question definitively, individual patient records are needed. Three large datasets exist that could provide them:

- **N3C** (National COVID Cohort Collaborative): 15 million patients, 75+ hospital sites, full EHR data including hospital site identifiers. Individual-level analysis within N3C would directly test whether neurological involvement predicts severity after patient-level adjustment for age, sex, and comorbidities.
- **ISARIC**: 705,000 hospitalized patients, 1,500+ sites, 60 countries. The international scope makes it the closest analogue to 4CE’s design.
- **Federated re-analysis of 4CE itself**: Sites run a standardized analysis script locally and share only coefficients. This preserves patient privacy while enabling individual-level estimation.

3.2 For multi-site studies in general

The ecological fallacy is not specific to COVID-19 or to the 4CE consortium. Any multi-site study that pools site-level summaries faces the same risk. Genome-wide association studies, multi-center drug trials, educational interventions across school districts, and crime statistics across cities—all are vulnerable when the units of analysis (sites) differ systematically from the units of inference (people).

Our results suggest five practical guidelines:

1. **Report heterogeneity honestly.** $I^2 = 99.8\%$ means the pooled average is almost meaningless. When heterogeneity exceeds 90%, the meta-analytic average should be presented alongside the full distribution of site-level effects, with explicit acknowledgment that the average may not apply to any individual site.
2. **Test the structure, not just the average.** Consistency tests can detect when the relationships between variables differ across sites in coordinated ways. This is information that Cochran’s Q misses because Q tests one variable at a time, and that geometric distance methods miss because they detect expected demographic variation rather than unexpected relational heterogeneity.
3. **Use hierarchical models, not separate tests.** When running multiple tests across timepoints, subgroups, or methods, the standard fix is Bonferroni or FDR correction. A better approach is to fit a single hierarchical model with site as a random effect and time as a covariate, so that partial pooling automatically shrinks noisy estimates toward the group mean. This addresses the multiplicity problem at the model level rather than the p -value level, and directly tackles the ecological-inflation problem: implausibly large effect estimates from few sites are likely Type M (magnitude) errors that shrinkage corrects.
4. **Disclose the full analysis menu.** The cheapest integrity fix for any study running multiple methods is to state explicitly how many tests were conducted and report all of them, including null results. Undisclosed analytic flexibility—not the raw count of tests—is what destroys credibility.
5. **Seek individual data when possible.** The ecological fallacy is a problem of aggregation, and the only definitive solution is disaggregation. Modern data-sharing frameworks (federated analysis, secure enclaves,

synthetic data) make individual-level analysis feasible even when raw data cannot leave the hospital. The convenience of site-level summaries does not justify the risk of reaching the wrong conclusion.

4 Methods

4.1 Data sources

CDC Case Surveillance. The CDC COVID-19 Case Surveillance Public Use Dataset contains 38,173,454 confirmed and probable COVID-19 cases reported to the CDC through 2022. Each record includes onset date, age group (9 categories: 0–9 through 80+), sex, race/ethnicity (9 categories), hospitalization, ICU admission, death, and underlying medical condition (binary flag). No geographic identifiers are present in the public version. We used race/ethnicity categories as pseudo-sites, restricted to groups with $n \geq 1,000$ (9 groups, n ranging from 13,548 to 6,165,036).

Zenodo New York Hospitals. A publicly available dataset of 1,540 COVID-19 patients from Maimonides Medical Center and SUNY Downstate Medical Center in Brooklyn, New York. Records include age (continuous), sex, race, outcome (expired/discharged), hospital identifier, and 13 binary comorbidity flags (hypertension, diabetes, COPD, CHF, etc.).

4CE Consortium. Aggregate site-level data from the 4CE analysis of neurological COVID-19 outcomes: 21 healthcare sites, 6 countries, over 22,000 hospitalized patients, with ICD-10 coded neurological diagnoses and severity outcomes at 30, 60, and 90 days Chou et al. (2021).

4.2 Statistical analyses

Ecological regression. For each pseudo-site (race/ethnicity group in CDC data), we computed the proportion of patients aged 80+ and the overall mortality rate. We regressed site mortality on site elderly proportion using ordinary least squares across the 9 sites.

Individual-level logistic regression. We fit logistic regression models on the complete individual-level data: (1) death $\sim$ age; (2) death $\sim$ age + sex + comorbidity; (3) death $\sim$ age $\times$ comorbidity + sex. For the CDC data, age was coded as the midpoint of the age group (5, 15, ..., 85). Odds ratios are reported per decade of age. We used `statsmodels` v0.14 for estimation.

Per-site correlation matrices. For each pseudo-site, we computed the Pearson correlation matrix among six binary variables: death, hospitalization, ICU admission, comorbidity, male sex, and age ≥ 80 . For the Zenodo data, we computed correlations among death and 13 comorbidities within each hospital.

Multivariate consistency test. Inspired by sheaf cohomology in sensor integration Robinson (2017), we measure whether site-level correlation structures are globally consistent. We do not compute a formal coboundary operator or cohomology class; rather, we flatten each site’s correlation matrix to its upper triangle (15 values for 6 variables) and compute the total inconsistency as the mean sum of squared pairwise differences across all site pairs. Significance was assessed by permutation: for each of 500 iterations, we shuffled each correlation value independently across sites (preserving the marginal distribution of each pair) and recomputed the inconsistency. The z -score compares the observed inconsistency to this null distribution.

Interaction heterogeneity test. For each pseudo-site, we fit a logistic regression with interaction terms (age $\times$ comorbidity, age $\times$ sex, comorbidity $\times$ sex) and extracted the interaction coefficients. The heterogeneity norm is the Frobenius norm of the cross-site deviation in these coefficients from their mean. Significance was assessed by permutation (500 iterations, shuffling interaction coefficients across sites within each interaction type).

Cochran’s Q. For comparison, we computed Cochran’s Q on each individual correlation pair across sites, using Fisher’s z -transform for variance stabilization.

Information geometry (Fisher–Rao distance). Each site defines a point on the product manifold of Bernoulli distributions (one Bernoulli per binary variable). The Fisher–Rao distance between two Bernoulli distributions with parameters p_1, p_2 is $d_{FR} = 2|\arcsin \sqrt{p_1} - \arcsin \sqrt{p_2}|$, the geodesic distance on the sta-

tistical manifold. For the product manifold, the total distance is the root sum of squares of per-variable geodesic distances. The Fréchet mean (the point minimizing total squared geodesic distance to all sites) is computed via the arcsine-transformed mean. Geodesic spread (the mean squared Fisher–Rao distance from the Fréchet mean) quantifies geometric heterogeneity. Significance was assessed by permuting variable proportions independently across sites (500 iterations).

Optimal transport (Wasserstein distance). For product Bernoulli distributions, the 1-Wasserstein distance is the sum of marginal absolute differences: $W_1(p, q) = \sum_j |p_j - q_j|$. The Wasserstein barycenter (population-weighted) gives a principled “average site” that minimizes total transport cost. The total cost measures how much work is needed to align the sites. Significance was assessed by permutation (500 iterations).

4CE meta-analysis. Fixed-effects, random-effects (DerSimonian–Laird), and influence-function meta-analysis of the proportional morbidity index (PMI) for CNS neurological involvement on COVID-19 severity, as described in our companion analysis Chou et al. (2021). Meta-regression of log-PMI on four site-level demographics (proportion female, proportion age ≥ 80 , baseline severity, number of patients).

4.3 Reproducibility

All code and data are available at [\[REPOSITORY URL\]](#). The CDC data is publicly downloadable from [data.cdc.gov](#) (dataset `vbim-akqf`). The Zenodo data is at [\[ZENODO DOI\]](#). Analysis scripts are in [experiments/visweswaran/data/](#).

5 Limitations

Pseudo-sites are not hospitals. Race/ethnicity groups in the CDC data are a proxy for the multi-site structure of consortium studies. They differ from hospitals in important ways: they are not geographically localized, patients within a group may attend different hospitals, and the grouping confounds race with socioeconomic factors. The ecological fallacy demonstration is valid—group-level analysis misses patient-level effects—but the specific confounders differ from those in a hospital-based consortium.

Limited variables. The CDC public dataset has 12 variables. Hospital-based datasets like N3C have hundreds. Our analysis captures the direction of the problem but underestimates its magnitude: with more variables, the scope for ecological confounding grows.

Two-hospital Zenodo data. With only two hospitals, the Zenodo data cannot support geometric methods (which need many sites) or detect interaction heterogeneity. It serves as a patient-level cross-check, not a standalone analysis.

Consistency tests are not a substitute. The multivariate consistency test and interaction heterogeneity test detect *that* something is wrong with site-level pooling; they do not tell you *what* the individual-level answer is. They are diagnostic tools, not corrective ones.

Geometric distance methods detect the wrong kind of heterogeneity. Information geometry and optimal transport measure whether sites differ in their marginal distributions (e.g., different proportions of elderly patients). This is expected demographic variation and does not signal a problem with pooling. The consistency test detects relational heterogeneity (do variables *relate* differently across sites?), which directly undermines pooling. The null results from the geometric distance methods are informative: they confirm that the consistency test is detecting something beyond demographic variation.

Multiple testing. We ran seven methods on the same data. In the companion analysis paper, we ran additional methods across multiple timepoints and subgroups. We report all results, including null findings, and apply FDR correction where applicable. Some findings that are individually significant may not survive correction. We interpret this as consistent with the paper’s thesis: site-level signals are often weaker than they appear.

6 Conclusion

A pooled average across hospitals can obscure the truth about individual patients. We showed this directly: a site-level analysis of 38 million COVID-19 cases cannot detect the relationship between age and mortality, while individual-level analysis finds a 9.9-fold odds ratio. The same problem likely affects the 4CE consortium’s neurological findings, where extreme heterogeneity ($I^2 = 99.8\%$) and strong dependence on site demographics create the conditions for ecological confounding.

Structural consistency tests offer a partial solution. They cannot replace individual-level analysis, but they can detect when site-level data is structurally inconsistent—a warning that the pooled average should not be trusted. For the 4CE data, these tests reveal coordinated heterogeneity in comorbidity correlation patterns ($z = 25$, $p < 0.0001$) that standard tests miss entirely.

The practical recommendation is simple: when heterogeneity is extreme, do not trust the average. Seek individual data. When individual data is unavailable, test the structure. And when reporting pooled results, say clearly what the average can and cannot tell you.

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